Satyam Shivam Sahu

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TECHNICAL SKILLS

Languages: Python, SQL, C, C++, Bash.
AI & Agentic Systems: LangGraph, LangChain, MCP (Model Context Protocol), Prompt Defense, LLM Guardrails, Sentence Embeddings, LangSmith.
Backend & Architecture: FastAPI, SQLAlchemy, Celery, Alembic, Tenacity, Pytest, Pydantic, REST APIs, ETL, Group Relative Policy Optimization, Direct Preference Optimization, Odds Ratio Preference Optimization.
Cloud, Data & Ops: PostgreSQL (pgvector, RLS), Redis. MongoDB, AWS (EC2, S3, RDS, Lambda), Docker, CI/CD GitHub Actions.

PROFESSIONAL EXPERIENCE

APPSOURCE SERVICE PVT LTD | Bengaluru, IND **Nov 2024 – Present**
Python Developer

- Prevented cross-tenant data leakage at the query layer by developing an end-to-end document intelligence platform using FastAPI and LangGraph, with strict PostgreSQL domain isolation.
- Delivered sub-second semantic retrieval across enterprise document repositories by designing a hybrid vector and Reciprocal Rank Fusion (RRF) search pipeline powered by pgvector.
- Blocked prompt injection attacks in production workflows by integrating an inline Grader Node firewall into the LangGraph execution graph to sanitize user prompts before LLM inference.
- Ensured zero data egress for sensitive enterprise records by running localized Llama 3.1 8B inference pipelines directly inside private infrastructure.
- Reduced document indexing turnaround time to under 3 seconds per file by creating an automated ingestion pipeline with PyMuPDF and Hugging Face sentence embeddings.
- Scaled the end-to-end data pipeline to handle 21,000 records/sec, while confirming fault tolerance via automated failure-injection tests that simulate message broker crashes.
- Ensured idempotent writes and at-least-once task delivery during worker outages by building a Celery execution engine featuring distributed batch fencing and a transactional outbox.
- Removed third-party API sync failures triggered by provider rate limits by developing an extensible Python connector framework with exponential backoff and retry logic.
- Enforced tenant data boundaries at the database engine level by applying PostgreSQL Row-Level Security (RLS) to both scheduled and on-demand ingestion workflows.
- Secured external HTTP data connectors against Server-Side Request Forgery (SSRF) vulnerabilities to safeguard internal cloud network infrastructure.

FINETUNING LLMs

- Group Relative Policy Optimization (GRPO): Reinforcement learning based fine-tuning of SmolLM-135M using TRL's GRPOTrainer and LoRA ($r = 8, \alpha = 16$). Removed the need for a separate critic/value model altogether by finding policy updates against a group-sampled baseline across 4 generations per prompt. Authoring deterministic reward functions for factual correctness via token overlap and SequenceMatcher, structural clarity, and conversational help.
- Direct Preference Optimization (DPO): Alignment of Qwen 2.5 (0.5B) using 4-bit QLoRA (NF4 double quantization, $r = 16, \alpha = 32$) in conjunction with TRL's DPOTrainer and a sigmoid loss function ($\beta = 0.1$), converted preference pairs into chat templates to penalize rejected outputs and push evaluation reward margins to +1.21 without incurring out-of-memory errors on commodity GPU hardware .

- Odds Ratio Preference Optimization (ORPO): Fine-tuning of Qwen2 (0.5B) on UltraFeedback preference data using TRL's ORPOTrainer in combination with bfloat16 mixed precision and LoRA ($r = 8, \alpha = 16$). Monolithic instruction-tuning and alignment in one training phase with no reference model with gradient checkpointing to enforce sub-512 token context windows .

PROJECTS

TripMate AI | Python, LangGraph, FastAPI, Streamlit, PostgreSQL, FastMCP, LangSmith

- Designed and built TripMate AI, an end-to-end multi-agent travel planner orchestrating specialized agents (Flight, Hotel, Weather, Budget, and Itinerary) using LangGraph to manage complex, parallel workflows. Architected state persistence and checkpointing by integrating a remote PostgreSQL database running on Render cloud.
- Implemented an LLM-driven Supervisor Agent (Llama 3.3 via Groq) to dynamically evaluate user intent and trigger only the necessary specialized agents for a given query, eliminating redundant sequential operations and drastically optimizing API and LLM token costs.
- Replaced brittle, hardcoded API endpoints with a standardized Model Context Protocol (MCP) layer, integrating remote (Tavly), local (Aviation Stack), and custom-built (OpenWeather) MCP servers. This decoupled the core agent logic from third-party schemas, establishing a self-healing architecture resilient to downstream API migrations and updates.
- Built a robust model-based guardrail system at the ingestion layer using Llama 3.3 to analyze user prompts for safety and domain relevance, automatically blocking harmful, malicious, or out-of-scope requests before triggering agent workflows.
- Designed an interactive Human-in-the-Loop (HITL) workflow using LangGraph's state interruption and command features, pausing execution to allow users to review draft itineraries, submit modification feedback, and approve revisions before compiling the final travel plan into a downloadable.

EDUCATION

O.P. JINDAL UNIVERSITY **2019 – 2023**
Bachelor of Technology in Computer Science and Engineering

CERTIFICATION

Oracle Cloud Infrastructure Certified Architect Associate, ID: 330825704OCI26CAA
Oracle Agentic AI Certified Associate, ID: 330825704AAI26OFA
Software Engineer Certificate by HackerRank, ID: db0fbcb3d522
SQL (Advanced) Certificate by HackerRank, ID: c111e1c56385
ML, DL and Data Science with Python, ID: 73558C4534-73555074B2-7355504E04
Walmart USA Advanced Software Engineering, ID: vktxj2P54d4SZfE4p
Deloitte Australia Technology, ID: oezMsAqtpEhu86FSa